Higher and Increasingly Higher Relative HIV Risk for Women: Evidence from Nearly One Million Blood Tests

Christian Aleman [1] Daniela Iorio [2] Raül Santaeulàlia-Llopis [3]

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[1]NYU Abu Dhabi

[2]Università di Bologna

[3]NYU Abu Dhabi, Barcelona SE and CEPR

Research Question

- What is the HIV prevalence for women vs. that for men in Sub-Saharan Africa?
- How does the gender HIV prevalence gap evolve over the **stages** of the epidemic?
- How much of the gender HIV prevalence gap is explained by age, marital status, area of living, education?

What we do

We characterize the evolution of the aggregate HIV prevalence in a comparable manner across countries and time: Define **stages** of the HIV epidemic (Aleman et al., 2023a).

(1) Assemble a large data set from 735K individual HIV blood tests in SSA: 30 countries, 60 country-year surveys (all DHS surveys with HIV data).

(2) Document the evolution of the gender HIV prevalence gap across **stages**.
$\longrightarrow$ Explore it by age, marital status, residency, education...

What we don't do (but do somewhere else...)

(1) We do not endogenize the epidemic (see Aleman et al., 2023a).

(2) We do not address the effects of policy neither theoretically (see Aleman et al., 2023a) nor empirically (proposed a new methodology in Aleman et al., 2023b)

What we find

(1) The gender HIV prevalence gap significantly increased from **53%** at relatively early stages (around the peak) of the epidemic to **87%** at more advanced stages.

(2) There is substantial heterogeneity in the gender HIV prevalence gap:

- within age groups:
  - it **decreased** from 137% to 40% for teenagers, ages 15-19 (drops to one third)
  - it **increased** from 80% to 175% for ages 20-34 (more than doubles)
  - it **increased** from 10% to 45% for ages 35-49 (increases four times)
- by marital status: it **decreased** from 150% to 70% for the never married, and **increased** from 5% to 39% for married.
- by area of living: it **increased** from 60% to 110% in urban areas. In rural areas from 40% to 70% .
- By years of education: it **increased** from 35% to 85% for people with primary or less (yeduc $<=6$), and from 75% to 125% for people more than primary (>6)

Related Literature

- Young women at risk through inter-generational sex, condom choice disability, gender norms (Dupas 2011), (Tsai et al. 2012), (Anderson et al. 2011), (Abeler et al.2023).
- Relationship between HIV, risky sexual behavior, and HIV knowledge , (DeWalque 2007), (Delavande, Kohler 2012), (Greenwood Kircher Tertilt 2013).
- HIV-Education gradient (Hargreaves Glynn 2002), (Wojcicki 2005), (Strauss Thomas 2007), (Beegle-DeWalque 2009) and (Aleman, Iorio, Santaeulalia 2023).
- Stage based analysis of aggregate outcomes (Galor, Weil 2000), (Lucas 2004), (Iorio-Santaeulalia-2011).

Outline

The Data

DHS Sample:
60 country-year surveys, 30 countries, 735K individual HIV blood tests

Table 1: HIV Prevalence by subgroups (%) ‣ Country year table

	Overall	13-45 (87.0%)
Overall (735 455)	5.93	6.08
Male (45%)	4.59	4.61
Female (55%)	7.05	7.20
Rural (66%)	4.95	5.09
Urban (34%)	7.83	7.92
Never sexually active (17%)	1.11	1.11
Sexually active (83%)	7.15	7.19
Primary or less (57.3%)	4.14	4.29
More than Primary (42.7%)	8.34	8.27
Married (58.8%)	6.18	6.25
Never Married (33.9%)	3.17	3.15

Definition of Stages

General Definition of Stages

Define a **stage** of a regional outcome at time t as its location on the support of a reference outcome path. e.g the following stylized path:

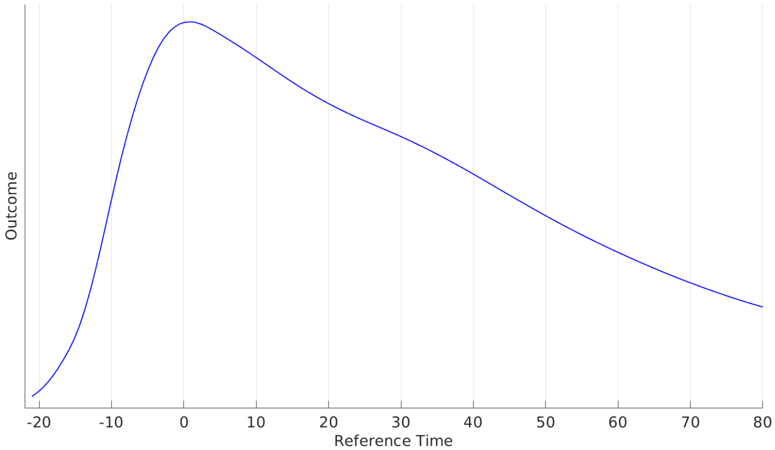
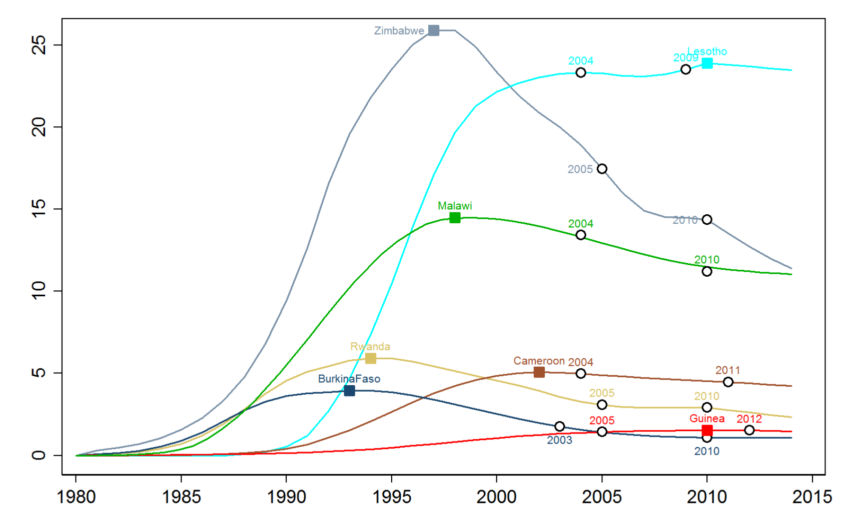


Figure 2: Challenges for the Definition of Stages of the HIV Epidemic.



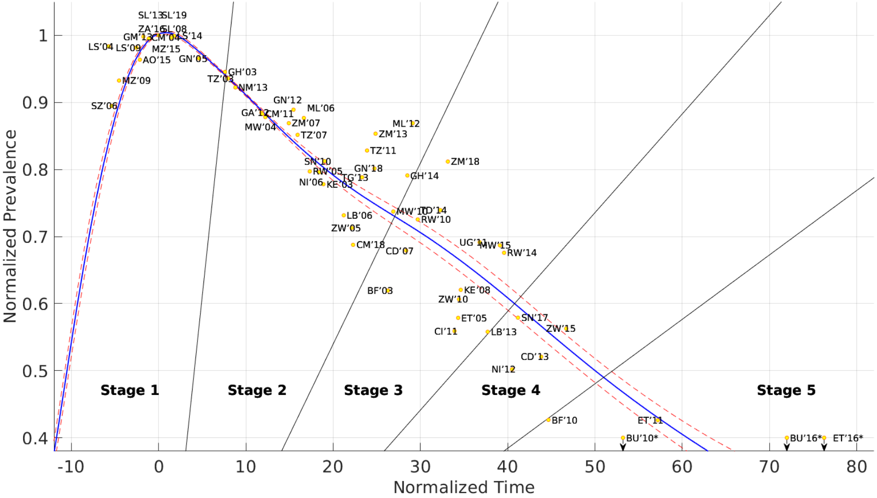
Challenges for the Definition of Stages of the HIV Epidemic

The HIV epidemic evolves differently across SSA countries:

- HIV prevalence at the peak,
- The year of the HIV peak,
- The time it takes each country to reach its own peak,
- The pace at which each country moves away from its peak.

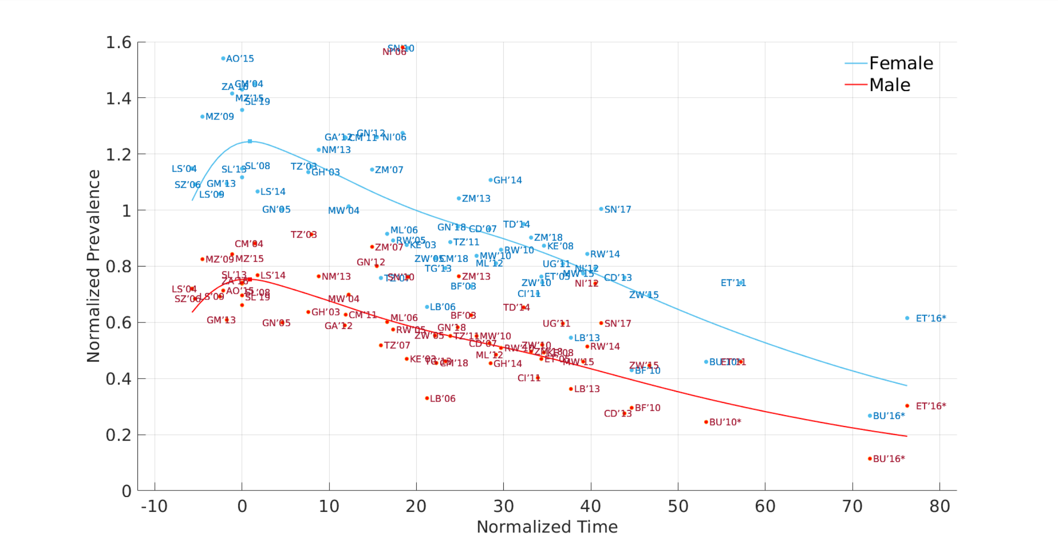
Solution: 2D normalization procedure in Iorio and Santaeulalia-Llopis (2023)

A Definition of the Stages of the HIV Epidemic: 2D Norm

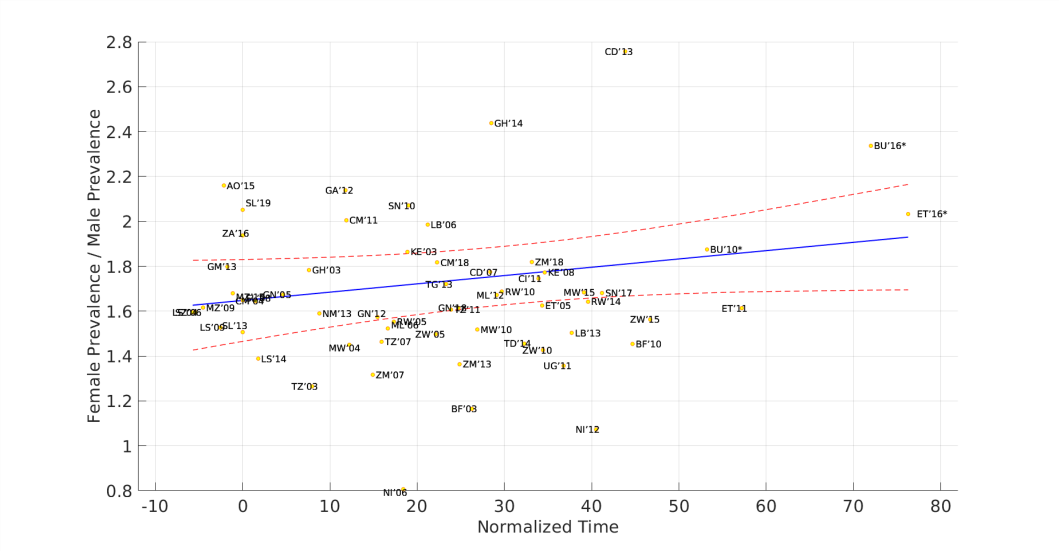


The Gender HIV Prevalence Gap

Evolution of the HIV prevalences by gender:

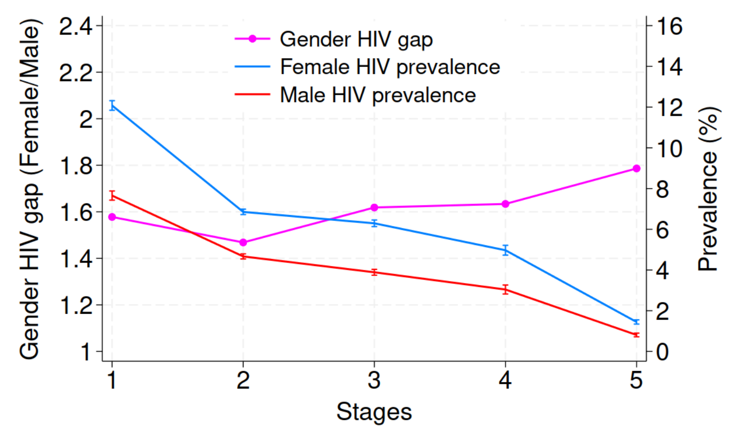


Evolution of the relative HIV prevalence gap:



Evolution of the relative HIV prevalence gap, discrete stages:

Figure 3: The HIV prevalence by gender and relative gender gap across **stages** for people aged 15-49.



A Statistical Model

We specify:

$$y_{i(t),t,j} = \alpha + \left(\gamma_1 + \sum_{j>1} \gamma_j \mathbf{1}_j\right) female_{i(t),t,j} + \beta\, x_{i(t),t} + \psi\, m_{c,t} + \theta_t \mathbf{1}_t + \theta_c \mathbf{1}_c + \varepsilon_{i(t),t}, \tag{1}$$

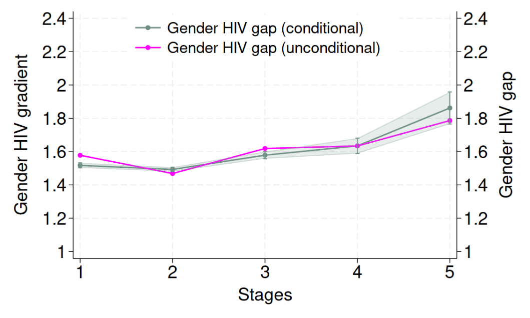
The course of the epidemic can be defined by **stages** $(j) \in \{1, 2, 3, 4, 5\}$

- $y_{i,t,j}$ being **HIV status**.
- *female* (binary) records the gender of the individual.
- $x_{i(t),t}$ includes age, age squared, urban rural (education, marital status).
- $m_{c,t}$, corrects for the *stage* of economic development in which each country. We use agricultural share and GDP pc.
- θ_t, θ_c are year and country dummies

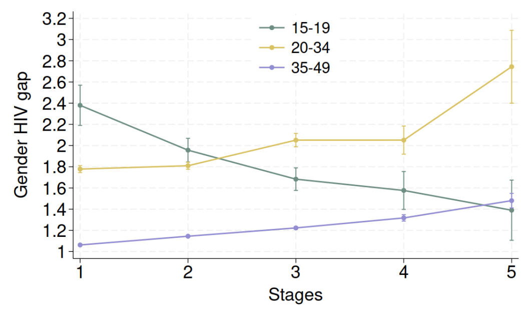
Our parameter of interest is γ_j.

Evolution of HIV Prevalence Gap, by age groups ▸ Tables

(a) Age 15-49

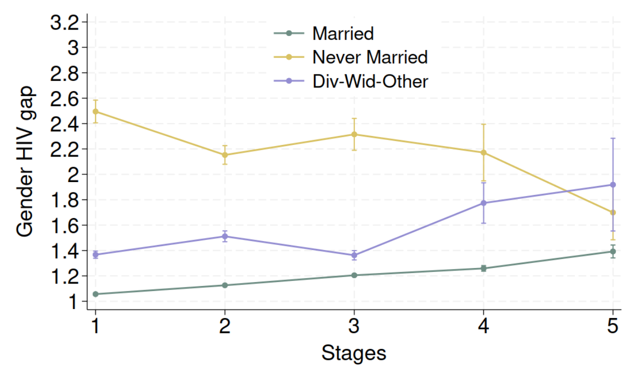


(b) All ages



Evolution of HIV Prevalence Gap, by marital status

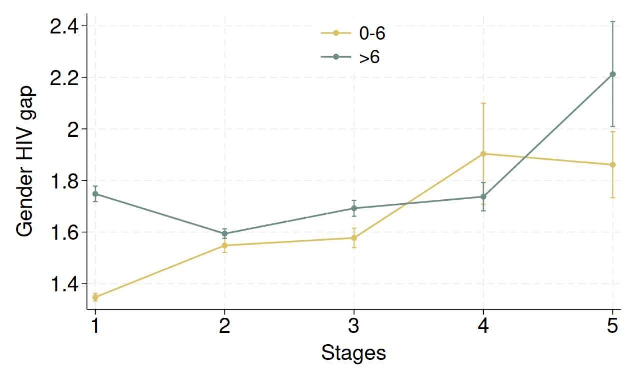
(c) By Marital Status



% Married	55.17	57.52	57.38	60.57	59.60
% Never Married	37.51	35.53	34.87	33.17	34.31

Evolution of HIV Prevalence Gap, by years of education

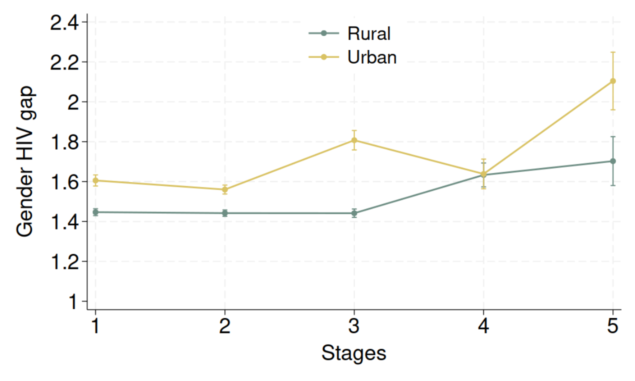
(d) By years of Education



% 0-6	53.67	50.02	56.23	48.28	73.09
% > 6	46.33	49.98	43.77	51.72	26.91

Evolution of HIV Prevalence Gap, by area of living

(e) By area of living



% Urban	41.74	36.64	29.17	38.30	28.58
% Rural	58.26	63.36	70.83	61.70	71.42

Conclusions

Conclusions

- The **gender HIV prevalence gap increases** over the stages of the of the epidemic.
- What's behind the increase in the gender gap?

▷ Age: declining gap for teenagers and **increasing gap for adults.**

▷ Marital status: declining gap for never married and **increasing gap for married**

▷ No major role for education or area of living (gap increases in all cases).

Targeted Policy

- Is this the right direction?

The Economist

Menu

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Leaders | How to beat HIV

To end AIDS, high-risk countries will need to jab schoolgirls

Injections that could keep a generation virus-free are on the horizon

Q&A

A Definition of the Stages of the HIV Epidemic

- Goal: Map each country$\times$year observation into a stylized (average) HIV epidemic.
- How: A normalization of the epidemic across time and space:
  (1) Level Normalization
  (2) Time Normalization
- Then, we can readily pin down the HIV epidemic stage of each country$\times$year observation (i.e., each DHS dataset).
- We follow the same approach previously used to stylize the stages of the demographic transition and the stages of development.

A Normalization of the Evolution of the HIV Epidemic

Algorithm 1. *Given a time series of HIV prevalence of country i:*

(1) Interpolate the country-specific time path of prevalence for each country i. Interpoland:

$$s_i(t) = \sum_{j=1}^{n} \theta_j \, \psi_j(t)$$

with n unknown θ_j coefficients. Denote the aggregate interpoland as $s(t)$.

(2) Level normalization

1. *Compute the country-specific peak prevalence,* $s_i(t_*^i) = \max_t s_i(t)$ *where* $t_*^i = \arg\max_t s_i(t)$ *is the period country i reaches its peak,* $s_i(t_*^i)$.
2. *Normalize the country-specific and aggregate interpolands by their respective peak prevalence,*

$$\widetilde{s}_i(t) = \frac{1}{s_i(t_*^i)} \, s_i(t) \quad \textit{and} \quad \widetilde{s}(t) = \frac{1}{s(t_*)} \, s(t),$$

Time normalization

1. *For $t_0^i < t^i \leq t_*^i$, normalize the time interval between the initial period for which data are available, $t_0^i = 1980$, and the country-specific peak period, t_*^i, by the time interval between the aggregate initial period, $t_0 = 1980$, and the aggregate peak period, t_*. To do so, we compute the constant of time proportionality for the pre-peak era,*

$$\alpha_i^L = \frac{t_* - t_0}{t_*^i - t_0^i}.$$

For $t^i > t_*^i$, normalize the time between the peak period, t_*^i, and the period t_γ^i in which country i reaches a given threshold $\gamma \in [0, 1]$,

$$\alpha_i^R(\gamma) = \frac{t_\gamma - t_*}{t_\gamma^i - t_*^i}.$$

2. *Normalize the time input of the country-specific interpolands by α_i^L and α_i^R,*

$$\tau = \alpha_i^L (t - t_*^i) \quad \textit{for} \quad t \leq t_*^i \tag{2}$$

$$\tau = \alpha_i^R (t - t_*^i) \quad \textit{for} \quad t > t_*^i \tag{3}$$

where $\tau \in T$ are the normalized units of time.

Data: DHS Sample

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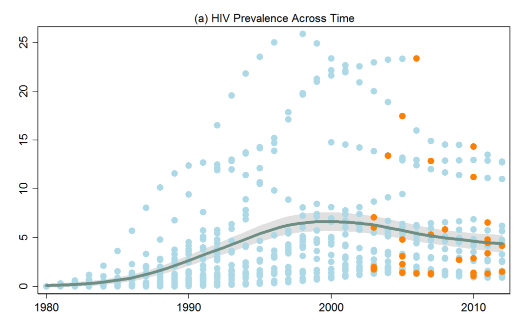
Table 2: DHS Sample

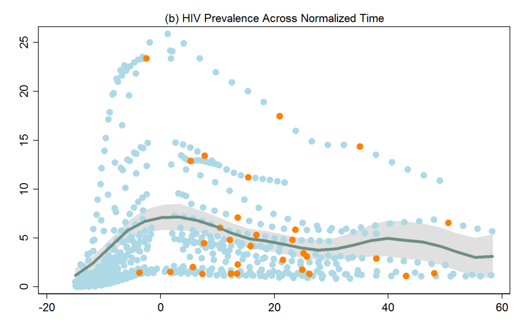
Country	Year(s) DHS Sample	Country	Year(s) DHS Sample
Malawi	2004,2010,2015	Gabon	2012
Burundi	2010,2016	Ghana	1998,2014
Congo DemRep	2007,2013	Kenya	2003,2008
Ethiopia	1997,2003,2008	Lesotho	2004,2009,2014
Guinea	2005,2012,2018	Liberia	2006,2013
Mozambique	2009,2015	Mali	2006,2012
Niger	2006,2012	Senegal	2010,2017
Rwanda	2005,2010,2014	Eswatini	2006
Sierra Leone	2008,2013	Uganda	2011
Tanzania	2003,2007,2011	Angola	2015
Zambia	2007,2013,2018	Chad	2014
Zimbabwe	2005,2010,2015	Gambia	2013
Burkina Faso	2003,2010	Namibia	2013
Cameroon	2004,2011,2018	South Africa	2016
Cote Ivoire	2011	Togo	2013

A Definition of the Stages of the HIV Epidemic ▸ Back

- Goal: Map each country$\times$year observation into a stylized (average) HIV epidemic.
- The stage of the epidemic is the result of a 2D normalization:
  (1) Level Normalization
  (2) Time Normalization

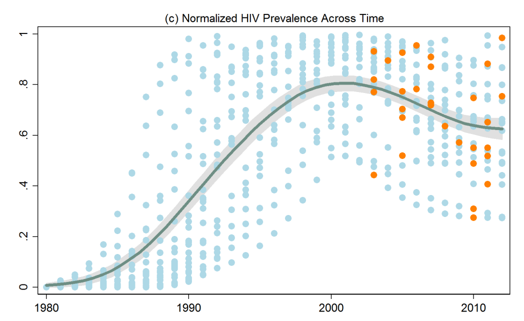
  $\longrightarrow$ We pin down the stage of the epidemic for each country$\times$year observation (i.e., each DHS dataset).

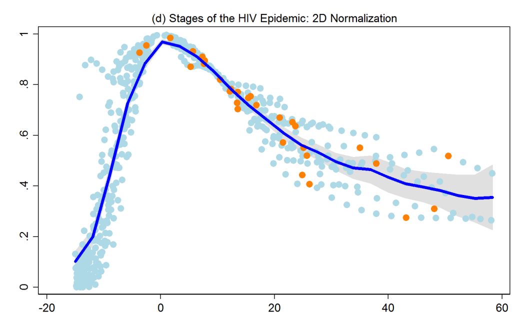




(c) Normalized Level

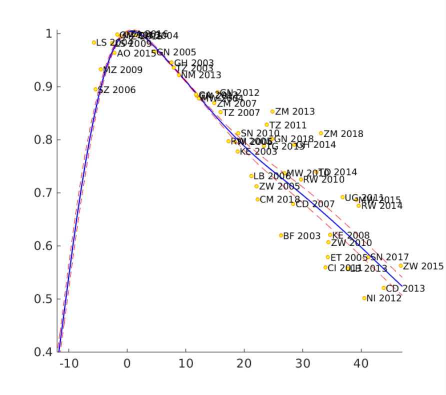
(d) Normalized HIV Epidemic



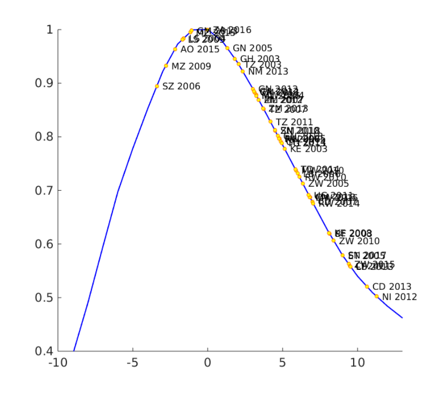


Stages of the HIV Epidemic ▸ Back

(a) Larger Sample



(b) Leading Region



Notes: Panel (a) shows the Normalization as described in the previous section. Panel (b) shows an alternative normalization using a leading country (Ethiopia) as reference.

Table 3: The linear combination for the HIV Gender gradient. ▸ Back

HIV-Gender gradient	(13-45)	(13-19)	(20-30)	(31-45)	(>45)	(<13)	(Never sexua active)
γ_1	0.042***	0.020***	0.070***	0.018***	0.002***	-0.001***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$\gamma_1 + \gamma_2$	0.024***	0.010***	0.034***	0.021***	0.001***	0.000***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$\gamma_1 + \gamma_3$	0.024***	0.006***	0.033***	0.025***	-0.010***	-0.000***	0.001***
	(0.000)	(0.0000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$\gamma_1 + \gamma_4$	0.020***	0.004***	0.020***	0.026***	-0.014***	0.000***	0.000***
	(0.000)	(0.0000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$\gamma_1 + \gamma_5$	0.007***	0.000***	0.008***	0.011***	-0.008***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

P-value of the Chi Sq. statistics for the interaction variables in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.